

Bridges: From Probability and Entropy to Branes, with Worked Examples

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Abstract

We construct the maps between five layers, probability, information geometry, entropy, couplings and symplectic branes, and test which structures they preserve. Entropy’s negative Hessian is the Fisher metric. Marginal constraints become Lagrangian branes as conormal bundles, and the graph of the entropy differential meets such a brane exactly at the maximum-entropy law of the constraint, transversally because the Fisher metric is nondegenerate. When no positive law meets the constraint the brane is empty. In a worked example with three bits this happens exactly when the pairwise disagreement reaches $2/3$, the contextual fraction is $1 - 3\varepsilon$ below the threshold, and the Fisher form of the fibre blows up as the threshold is approached. Because a cell of the filler vanishes there, joint samples detect the boundary at distance m from $O(1/m)$ samples, while pairwise-only samples need $O(1/m^2)$. On the toric side, entropy is an admissible symplectic potential on the toric manifold of every nondegenerate coupling polytope, without the facet hypothesis of earlier work, so couplings are the points of a positive real Lagrangian and the bases of torus branes. When the toric manifold is Fano, only the torus branes at critical points of the disc potential are nonzero objects; for 2×2 couplings this is the Fréchet midpoint, which is generally not the product coupling. Gluing with a fixed coupling acts on couplings linearly, and it preserves the lattice of the torus exactly when its kernel is deterministic or constant. So gluing with a non-constant stochastic kernel is not induced by a torus-equivariant correspondence, a defect related to, though not identical with, the defects of stochastic kernels found in other layers of the programme.

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*`check.bridges.py` (9 checks) verifies numerically the computable statements proved here, on examples and random instances, and every number in the worked examples. Propositions 4.2, 4.3 and the derived invariants in Proposition 6.1 rest on the cited literature; results cited from the other papers are checked there.

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1 Introduction

The programme of which this note is part works with five layers:

- probability distributions, which describe states and uncertainty;
- information geometry, which measures how distinguishable nearby states are;
- entropy, which supplies a potential and selects canonical distributions;
- coupling spaces, which describe compatible joint realizations [3, 1];
- symplectic branes and their Fukaya-type categories, which could organize admissible structures, transformations and their higher compositions [3, §§7, 10].

The task is to construct the maps between the layers and to find out which structures they preserve. Each bridge below comes with a proof and a worked example in which every number can be checked by hand; the one bridge where data enter, the detection of a breaking point, comes with a synthetic test.

Main results.

- (A) *Constraint branes* (theorem 3.1). The conormal bundle of a marginal constraint is a Lagrangian in the cotangent bundle of the open simplex. The graph of dH meets it in exactly one point, the maximum-entropy law of the constraint, transversally; it misses it exactly when no positive law satisfies the constraint.
- (B) *Three bits* (example 3.3). For pairwise disagreement $1 - \varepsilon$, a joint exists iff $\varepsilon \geq \frac{1}{3}$, a positive one iff $\frac{1}{3} < \varepsilon < 1$; the contextual fraction is $1 - 3\varepsilon$ below $\frac{1}{3}$; the Fisher form of the fibre is $\frac{8}{3\varepsilon-1} + \frac{24}{1-\varepsilon}$; and from joint samples the threshold is detected at distance m with $O(1/m)$ samples.
- (C) *Toric branes* (theorem 4.1 and propositions 4.2 and 4.3). On the toric manifold of every nondegenerate coupling polytope, entropy is an admissible symplectic potential; couplings are the points of the positive real Lagrangian and the bases of the torus branes; when the manifold is Fano, only torus branes at critical points of the disc potential survive, which for 2×2 couplings is the Fréchet midpoint (example 4.4).
- (D) *Correspondences* (theorem 5.1 and example 5.2). Gluing with a coupling whose kernel is K maps lattice to lattice iff K is deterministic or constant, so a non-constant stochastic K is not induced by an equivariant correspondence; in the 2×2 case it scales the moment segment by the Dobrushin coefficient.
- (E) *Categorical layer* (proposition 6.1). The gluing algebra is Morita equivalent to $\mathbb{R} \times \mathbb{R}$, so all its derived invariants are those of two points.

What is not claimed. We do not construct a functor from couplings to a Fukaya category. We show where the obvious candidates fail and why: in the cotangent bundle of the simplex, which is contractible, the entropy brane carries only its one intersection point; in the Fano case, torus branes over most couplings are zero objects; and gluing with a non-constant stochastic kernel is

not induced by an equivariant correspondence of the kind defined in [section 5](#). Non-equivariant correspondences are not ruled out.

2 From probability to geometry

Let Δ° be the open simplex of full-support distributions on n points, with tangent space $T = \{u : \sum u_y = 0\}$, and $H(p) = -\sum p_y \log p_y$.

Proposition 2.1. *For $u, v \in T$, $-d^2 H_p(u, v) = \sum_y u_y v_y / p_y$, the Fisher metric.*

Proof. $\partial^2 H / \partial p_y \partial p_z = -\delta_{yz} / p_y$. □

The Fisher metric is symmetric, so it is not a symplectic form; the two structures coexist in the cotangent bundle, the metric on the base and the canonical form $\omega = \sum dp_y \wedge d\xi_y$ on $T^* \Delta^\circ$, and on the toric manifolds of [section 4](#), where the Fisher metric is twice the Kähler metric on the real section [[3](#), Prop. 6.5].

3 Constraint branes in the cotangent bundle

A marginal constraint fixes a linear image $Mp = b$ of the law. Its *fibre* is $C = \{p \in \Delta : Mp = b\}$ and its open part is $C^\circ = C \cap \Delta^\circ$. The fibre is not Lagrangian, but its conormal bundle is, as for every submanifold [[13](#)]:

$$N^* C^\circ = \{(p, \xi) \in T^* \Delta^\circ : p \in C^\circ, \xi|_{T_p C} = 0\}, \quad T_p C = \ker M \cap T.$$

The *entropy brane* is the graph $L_H = \{(p, dH_p)\}$, an exact Lagrangian.

Theorem 3.1 (Conormal intersection). *If $C^\circ \neq \emptyset$, then $L_H \cap N^* C^\circ$ is the single point over the maximum-entropy law p^* of C , and the intersection is transversal. If $C^\circ = \emptyset$, then $N^* C^\circ = \emptyset$ and the intersection is empty, although C itself may be nonempty.*

Proof. A point (p, dH_p) lies in $N^* C^\circ$ iff dH_p vanishes on $T_p C$, that is, iff p is a critical point of H on the relatively open convex set C° . H is strictly concave, so there is at most one such point, and it is the maximizer of H over C ; the maximizer lies in C° when $C^\circ \neq \emptyset$, because the derivative of H in a direction that raises a vanishing coordinate is $+\infty$. Transversality of the graph of dH to a conormal bundle at a point is nondegeneracy of the Hessian of $H|_C$ there, and by [proposition 2.1](#) that Hessian is minus the Fisher metric on $T_p C$, which is definite. □

Corollary 3.2. *Suppose the margins admit a strictly positive joint law, so that $C^\circ \neq \emptyset$. For the chain margins (X, Y) and (Y, Z) the intersection point is the conditionally independent gluing $P_{XY} P_{YZ} / P_Y$, the canonical filler of [[3](#), Thm. 4.2]. For pairwise margins of three variables it is the limit of iterative proportional fitting started at the uniform law [[7](#)].*

Proof. The gluing has the margins, and log of it is a sum of a function of (x, y) and one of (y, z) , which is orthogonal to the kernel of the marginal map. The second statement is the maximum-entropy characterization of fitting. □

So the brane picture of compatibility is exact: a constraint is compatible with positive laws iff its brane is nonempty, and then it meets the entropy brane exactly once, at the canonical filler.

Example 3.3 (Three bits). Let X_0, X_1, X_2 be bits, each uniform, and let each pair disagree with probability $1 - \varepsilon$, so that each pairwise table is $\begin{pmatrix} \varepsilon/2 & (1-\varepsilon)/2 \\ (1-\varepsilon)/2 & \varepsilon/2 \end{pmatrix}$.

- (i) *The fibre.* The kernel of the marginal map on $\mathbb{R}^8$ is spanned by the parity vector $x(s) = (-1)^{|s|}$, so the fibre is a segment along x .
- (ii) *Threshold.* Every assignment of three bits makes 0 or 2 of the three pairs disagree, so the expected number of disagreeing pairs, $3(1 - \varepsilon)$, is at most 2: a joint exists iff $\varepsilon \geq \frac{1}{3}$. At $\varepsilon = \frac{1}{3}$ the only joint puts no mass on 000 and 111, and at $\varepsilon = 1$ the only joint lives on 000 and 111; in both cases $C^\circ = \emptyset$ and the brane is empty. A positive joint exists iff $\frac{1}{3} < \varepsilon < 1$. The two ends are the breaking regime of [2].
- (iii) *The filler.* For $\frac{1}{3} < \varepsilon < 1$ the maximum-entropy law gives $a = \frac{3\varepsilon-1}{4}$ to 000 and 111 and $b = \frac{1-\varepsilon}{4}$ to the six other assignments, since then $\sum_s x(s) \log p(s) = 0$.
- (iv) *Transversality and its loss.* The Fisher form of the fibre at the filler is $\sum_s x(s)^2/p(s) = \frac{8}{3\varepsilon-1} + \frac{24}{1-\varepsilon}$, which is 64 at $\varepsilon = \frac{1}{2}$, 197 at 0.35 and 436 at 0.34, and diverges at both ends: the intersection becomes infinitely steep before it disappears.
- (v) *Contextual fraction.* Below the threshold the contextual fraction of [4] is $1 - 3\varepsilon$. A sub-normalized common joint λq below the tables puts at most ε on the agreements of each pair, and every assignment agrees on some pair, so $\lambda \leq 3\varepsilon$; the measure 3ε times the uniform law on the six assignments that agree on exactly one pair attains it.

Synthetic test: detecting the boundary. Take the positive law with $\varepsilon = \frac{1}{3} + m$ and N joint samples of the three bits, and estimate ε by the mean agreement rate. Each sample agrees on one pair, or on three if it is 000 or 111, so $\hat{\varepsilon} \leq \frac{1}{3}$ exactly when no all-equal sample appears, which happens with probability $(1 - \frac{3}{2}m)^N \approx e^{-3Nm/2}$. For $m = 0.02$ and $N = 25, 100, 400$ the verdict “no positive filler” was returned in 47%, 6% and 0% of 2000 replicates, against 46.7%, 4.8% and 0% predicted. So $O(1/m)$ joint samples suffice. The reason is that the filler’s cells on 000 and 111 vanish at the boundary: the Fisher information of ε in the joint model, $\frac{9}{2(3\varepsilon-1)} + \frac{3}{2(1-\varepsilon)}$, diverges there, and the test reduces to seeing a rare cell.

The rate depends on the design. In the usual contextuality design each pair is observed in its own samples, the agreement count of each pair is binomial, its Fisher information stays bounded at the boundary, and the misses decay only once Nm^2 is large: 42%, 26%, 7% and 0.05% for $N = 25, 100, 400, 1600$. Interior walls behave in the same way; for them [3, Prop. 12.2] shows that N of order $1/m^2$ suffices.

4 Toric branes

For nondegenerate margins f, g , the coupling polytope $\text{Cpl}(f, g)$ is Delzant for the lattice of integer circulations, and it is the moment polytope of a compact toric manifold $X_{f,g}$ [3, Thm. 6.4], [8]. The fibre of the moment map over an interior coupling is a Lagrangian torus. A symplectic potential on the polytope determines a compatible toric Kähler structure when it satisfies Abreu’s conditions [5]: it differs from Guillemin’s potential $G_P = \frac{1}{2} \sum_{\text{facets}} \ell \log \ell$ [11] by a function smooth on the closed polytope, its Hessian is positive definite inside, and $\det(\text{Hess } G) \prod_{\text{facets}} \ell$ is smooth and positive on the closed polytope.

Theorem 4.1 (Entropy is an admissible potential). *For nondegenerate margins, every cell α_c is either a facet function of $\text{Cpl}(f, g)$ or strictly positive on it, and $G = \frac{1}{2} \sum_c \alpha_c \log \alpha_c = -\frac{1}{2} H$ satisfies Abreu's conditions. Hence it defines a compatible toric Kähler structure on $X_{f,g}$ whose metric on the real section is half the Fisher metric, and whose potential has the product coupling as its unique critical point.*

Proof. The polytope is simple, and at each vertex exactly $(n-1)(m-1)$ cells vanish, as many as the dimension. At a vertex of a simple polytope given by a system in which exactly d inequalities are tight, each tight inequality defines a facet. Every face on which $\alpha_c = 0$ contains a vertex where it is tight, so α_c is a facet function or never vanishes. In the second case $\alpha_c \log \alpha_c$ is smooth on the polytope, so $G - G_P$ is smooth. The Hessian is $\frac{1}{2} \sum_c u_c^2 / \alpha_c$ on circulations, which is positive definite. Near a vertex, in Delzant coordinates in which the facets through it are $x_i \geq 0$ (each cell function is primitive on the lattice of integer circulations, since a basic move takes the value 1 on it, and distinct cells give distinct facets off the walls), $\text{Hess } G = \frac{1}{2} X^{-1} + S$ with $X = \text{diag}(x)$ and S smooth and positive semidefinite, so $\det(\text{Hess } G) \prod x_i = \det(\frac{1}{2} I + XS)$ is smooth and at least 2^{-d} . A boundary point is covered by the coordinates of a vertex of its minimal face, in which the remaining facet functions are positive factors. The critical point is computed in [3, Prop. 6.5]. $\square$

This removes the hypothesis of [3, Prop. 6.5] that every cell be a facet. Guillemin's own potential is a different admissible potential, and when some cells are not facets its critical point may differ from the product coupling (example 4.4).

Proposition 4.2 (The positive real Lagrangian). *The positive real part $X_{f,g}^{>0}$, the fixed set of complex conjugation in the open torus orbit with positive coordinates, is a Lagrangian submanifold, and the moment map restricts to a homeomorphism of its closure onto $\text{Cpl}(f, g)$ [10]. It meets the torus brane over each interior coupling in exactly one point. So the couplings are the points of one Lagrangian, and also the intersection points of that Lagrangian with the torus branes.*

Proof. Complex conjugation is antiholomorphic and preserves the torus-invariant Kähler form up to sign, so its fixed set is Lagrangian. The homeomorphism is the classical statement about the nonnegative part of a projective toric variety; a torus fibre meets the positive part in the single point whose angle coordinates are all zero. $\square$

Proposition 4.3 (Which torus branes survive). *Suppose $X_{f,g}$ is Fano. The torus brane over a coupling u , with a unitary local system, has nonzero Floer cohomology iff u and the local system give a critical point of the disc potential $\mathfrak{PD}(u; y) = \sum_{\text{facets}} T^{\ell(u)} y^\nu$, the Cho–Oh formula for toric Fano manifolds [6, 9]. All other torus branes are zero objects of the Fukaya category.*

Example 4.4 (2×2 couplings). Let $f = (0.4, 0.6)$ and $g = (0.3, 0.7)$. With $t = \alpha_{11}$ the polytope is the segment $[0, 0.3]$, the cells are $(t, 0.4 - t, 0.3 - t, 0.3 + t)$, and $X_{f,g} = \mathbb{CP}^1$.

- (i) The facets are $t \geq 0$ and $0.3 - t \geq 0$; the cells $0.4 - t$ and $0.3 + t$ stay at least 0.1 and 0.3, as theorem 4.1 predicts.
- (ii) The entropy potential is critical at $t = 0.12$, the product coupling $f_1 g_1$; Guillemin's potential is critical at the midpoint $t = 0.15$. The two Kähler structures differ, although their symplectic forms agree.
- (iii) $\mathfrak{PD} = T^t y + T^{0.3-t} y^{-1}$ has a critical point with unitary y only when both terms have equal valuation, $t = 0.15$, with $y = \pm 1$. So the only nonzero torus brane is over the Fréchet midpoint $\frac{1}{2}(\max(0, f_1 + g_1 - 1) + \min(f_1, g_1))$, the circle that divides the sphere into equal areas. Here

the brane over the product coupling is a zero object, as in [3, Prop. 10.11]. The two can coincide: for $f = (0.4, 0.6)$ and $g = (0.5, 0.5)$ the segment is $[0, 0.4]$ and the product coupling 0.2 is also the midpoint and Guillemin's critical point.

5 Correspondences: where gluing stops being geometric

Gluing with a fixed $\beta \in \text{Cpl}(g, h)$ is the map $\alpha \mapsto \alpha \text{diag}(g)^{-1}\beta = \alpha K$, where $K = \text{diag}(g)^{-1}\beta$ is the kernel of β . It is linear, and it carries $\text{Cpl}(f, g)$ into $\text{Cpl}(f, h)$ with $h = K^\top g$. The tori of $X_{f,g}$ and $X_{f,h}$ have the integer circulations of the two shapes as weight lattices [3, Thm. 6.4]. Call a Lagrangian correspondence $\Gamma \subset X_{f,g}^- \times X_{f,h}$ *equivariant* if it is connected and invariant under the torus $T_{f,h}$, acting on $X_{f,h}$ and on $X_{f,g}$ through a homomorphism $\psi: T_{f,h} \rightarrow T_{f,g}$, and say that it *induces* a map Φ of polytopes if $(\mu(z), \mu(z')) = (\alpha, \Phi(\alpha))$ for points $(z, z') \in \Gamma$ over a full-dimensional set of α . The moment map of the diagonal action is constant on the connected invariant Lagrangian Γ , so $\mu_{f,h} = d\psi^* \mu_{f,g} + c$ on Γ , and Φ has linear part $d\psi^*$, which maps weight lattice to weight lattice.

Theorem 5.1 (Integrality). *The linear map $x \mapsto xK$ carries integer $n \times m$ circulations to integer $n \times h$ circulations iff K is deterministic, every row a vertex of the simplex, or constant, all rows equal. So gluing with a coupling whose kernel is stochastic and not constant is not induced by any equivariant Lagrangian correspondence in the sense above.*

Proof. Integer circulations are generated by the moves $(e_i - e_k)(e_j - e_l)^\top$, and such a move goes to $(e_i - e_k)(K_j - K_l)$. So the condition is that any two rows of K differ by an integer vector. Two stochastic rows with integer difference are equal or are distinct vertices; if some row is not a vertex, every other row equals it. $\square$

Example 5.2 (2×2 gluing). For 2×2 couplings the circulations form the line spanned by $x = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, and $xK = \lambda x$ with $\lambda = K_{11} - K_{21}$. Its absolute value is the Dobrushin coefficient of K . The moment segment is carried onto a segment $|\lambda|$ times as long, so the map of moment polytopes contracts symplectic area by $|\lambda|$, and it is integral exactly for $\lambda \in \{1, -1, 0\}$: the identity, the swap and a constant kernel. In [example 4.4](#), a noisy relabelling with $K = \begin{pmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{pmatrix}$ has $\lambda = 0.7$ and shrinks the segment from length 0.3 to 0.21.

Remark 5.3 (Related defects). Stochastic kernels cause related, though not identical, defects elsewhere in the programme: whiskering is functorial for deterministic kernels and fails for stochastic ones in general [3, Thm. 10.2]; squares of couplings and kernels compose for bijections and sufficient merges, and fail for some non-injective functions and some constant kernels [1, Prop. 6.1]; and kernels contract bands by their Dobrushin coefficient [2]. The class singled out differs from case to case; here constant kernels are on the good side, there they are not. What the cases share is that the defect is a circulation, or a contraction of the circulation space.

6 The categorical layer

Couplings can enter a categorical layer as morphisms or as objects, and each choice loses something.

Proposition 6.1 (Morita data). *The gluing algebra $A_f \cong \mathbb{R} \times M_{n-1}(\mathbb{R})$ [1, Thm. 2.2] is Morita equivalent to $\mathbb{R} \times \mathbb{R}$. Hence its category of perfect complexes is that of two points, $K_0(A_f) = \mathbb{Z}^2$, its Hochschild cohomology is $\mathbb{R}^2$ in degree 0 (its centre, spanned by P_f and e_f) and zero above, and its simple modules have dimensions 1 and $n - 1$.*

Proof. $M_{n-1}(\mathbb{R})$ is Morita equivalent to $\mathbb{R}$, and Morita equivalence preserves derived categories, K -theory and Hochschild cohomology [12]. The centre of a product of matrix algebras is the product of the centres. $\square$

couplings as	composition	what survives
morphisms (spans)	exactly gluing [1]	for functors that kill e_f up to homotopy, only the mass [1, Cor. 5.3]; derived invariants of two points
torus branes	not equivariantly induced for non-constant stochastic kernels (theorem 5.1)	in the Fano case, only critical points of $\mathfrak{P}\mathfrak{D}$ (proposition 4.3)
points of $X_{f,g}^{>0}$	none	all couplings, with positivity (proposition 4.2)

Positivity, where contextuality lives [1, Cor. 5.3], is invisible in the first two rows: linear spans forget it, and which torus branes survive is decided by the disc potential, not by positivity. It is kept in the third row. So the promising structure is the Floer theory between the positive real Lagrangian and the torus branes.

7 Open problems

1. *Real and torus branes.* Compute $HF(X_{f,g}^{>0}, L_u)$, or of the full real locus, with its products, and decide whether the positivity of couplings survives in it.
2. *Non-equivariant correspondences.* Theorem 5.1 rules out equivariant ones. Is there a Lagrangian correspondence, not torus-equivariant, whose composition matches gluing with a stochastic kernel?
3. *Bulk deformations.* With bulk deformation the set of nonzero torus branes changes. Is there a bulk class for which the product coupling's brane is nonzero for all margins?
4. *Larger tables.* Locate the nonzero torus branes for $n \times m$ couplings, extending example 4.4; the computations of [3, Obs. 10.10] point to the leximin coupling.
5. *The three-bit brane under noise.* Track the conormal brane of example 3.3 as a family over ε , through the threshold, as a first example of a brane that disappears at a breaking point.

8 Computational checks

`check_bridges.py` runs nine checks, all passing: G1 (proposition 2.1); G2 (theorem 3.1 and corollary 3.2); W1 (example 3.3, all five parts); S1 (the synthetic detection test, with joint and with pairwise-only samples); G3 (theorem 4.1, the determinant condition at every vertex of an example with non-facet cells); W2 (example 4.4); G4 (theorem 5.1); W3 (example 5.2); W4 (proposition 6.1, the dimension and the centre).

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